

Review for Exam 1

Today Agenda

- Review for Exam 1
- Review Interrupt Hardware Component
- Example Practice

Reading

- **Slides week 1~ 3**
- **Introduction to FreeRTOS:**
 - Chapter 1. (Slides 1-87)
 - Chapter 7. (Slides 253-299)
- **FreeRTOS distribution:**
 - Sections 1.1-1.5: Task management
 - Sections 3.1-3.7: Priority and Timing.
 - Sections 7: Resource Management
- **Zynq Book Tutorials: (Slides 1-74)**
- **The ZYNQ Book Tutorials**
 - *1A, 1B, 1C, 2A, 2B, 2C*

Embedded System: Embedded SoC, SoC design flow, Zynq Board

- Reference Materials: Course lecture slide week 1 T&R, The Zynq Book Chapter 1~ 2, Zynq Book Slides 1~58
- Embedded system definition, Constraints of embedded system
- Advantage of FPGA
- SoC Architecture
- The definition and architecture of processing system in Zynq device
- Soft vs hard core
- APU Composition
- Architecture of PL: CLB
- Four layers of Zynq Platform
- Two performance metrics

FreeRTOS

- **Reference Materials:** Course lecture slide week 2 T&R, Introduction to FreeRTOS_Slides.pdf (1-87), FreeRTOS_Real_Time_Kernel-A_Hands-On_Tutorial_Guide.pdf (Chapter 1 and Chapter 3)
- Real-time system
- Types of real-time tasks
- What FreeRTOS does?
- Data Types in FreeRTOS: TickType_t, BaseType_t
- The definition of tick, tick_count
- The use of BaseType_t

Task Management

- **Reference Materials:** Course lecture slide week 2 T&R, Introduction to FreeRTOS_Slides.pdf (1-87), FreeRTOS_Real_Time_Kernel-A_Hands-On_Tutorial_Guide.pdf (Chapter 1 and Chapter 3)
- Define, create Task
- Suspend, resume Task
- Priority Set
- Time slice scheduling (per tick)
- pdMS_TO_TICKS
- Priority-based scheduling
- TaskDelay vs TaskDelayUntil

Hardware Interrupt

- **Reference Materials:** Course lecture slide week 2 T&R,
Introduction to FreeRTOS_Slides.pdf (1-87),
FreeRTOS_Real_Time_Kernel-A_Hands-On_Tutorial_Guide.pdf
(Chapter 1 and Chapter 3)
- Comparisons between polling and interrupt

Resource Management

- **Reference Materials:** Course lecture slide week 2 T&R, Introduction to FreeRTOS_Slides.pdf (1-87), FreeRTOS_Real_Time_Kernel-A_Hands-On_Tutorial_Guide.pdf (Chapter 1 and Chapter 3)
- Data consistency
- Critical section, mutex
- Priority inversion, deadlock
- Priority inheritance, time out
- Draw task execution diagram